



Address Spoofing Attacks and Defense

Technically, **address spoofing attacks** may be conducted using **IP address spoofing** or **MAC address spoofing** or a combination of both.

IP address spoofing is more commonly encountered in practice, e.g. using random IP addresses in TCP SYN flooding attack to be discussed in next Topic 2. Alternatively, IP address spoofing may be conducted by hijacking a valid IP address, possibly to compromise a target like the Smurf attack to be discussed in next Topic 2 as well, or in an attempt to bypass ACLs which will be discussed in this Subtopic.

Comparatively, **MAC address spoofing** is seldom heard of in practice as it is only feasible within an enterprise LAN. In addition, there exist better L2 attacks that were discussed in previous Subtopics. Nevertheless, as a network engineer in defending a network, it is essential to be aware of its existence and to be able to configure corresponding defense to counter it.

Thus, let's investigate two practical scenarios where these attacks are feasible.

Attack Scenario A: IP Address Spoofing Attack

It is common to implement ACLs in an enterprise LAN for access control. Consider the scenario where ACL is implemented at the distribution switch SW1 to only permit department manager Bob to access the server but not other staff as shown in Figure 1.

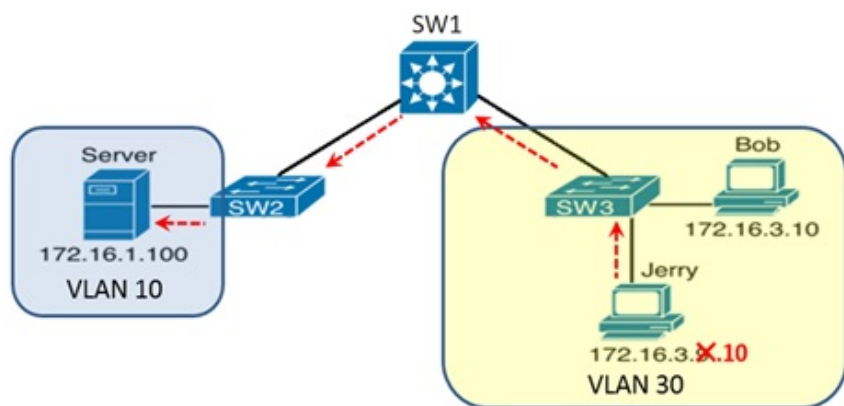


Figure 1

For simplicity, assume that the following ACL is implemented at SW1.

```
SW1(config)# access-list 1 permit 172.16.3.10
SW1(config)# int vlan 10
SW1(config-if)# ip access-group 1 out
```

Suppose that an insider threat actor Jerry is bribed by an adversary to access the server. What could he do?

Through prior reconnaissance and possibly sniffing of network traffic within the department, Jerry would know the IP address of Bob. Thus, a potential attack is for Jerry to conduct **IP address spoofing** by simply changing his PC IP address to that of Bob. In this case, Jerry could bypass the ACL to access the server.

Attack Scenario B: MAC Address Spoofing Attack

Consider a different scenario where another insider threat actor Jimmy happens to have access to VLAN 10 where the server is residing as shown in Figure 2.

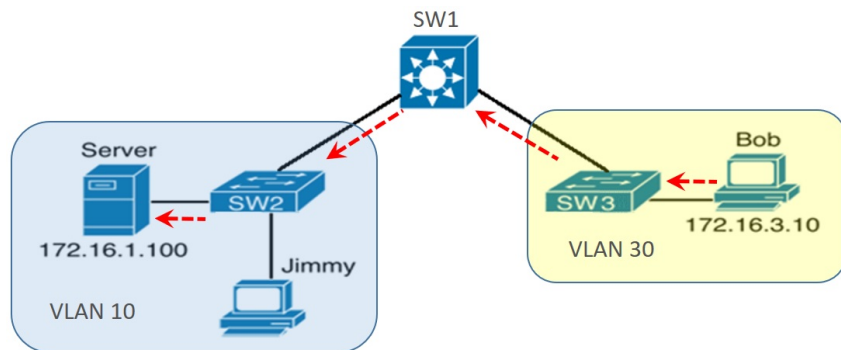


Figure 2

To steal the network traffic from Bob to the server, **MAC address spoofing** will be discussed in this scenario although better L2 attacks could be conducted.

Similarly, by prior reconnaissance and sniffing of the network traffic, Jimmy would know the MAC address of the server. To conduct **MAC address spoofing**, Jimmy will change his PC MAC address to that of the server. After that, he will need to send a frame out of his PC to update the MAC address table of access switch SW2 so that it will now forward all frames destined for the server to him as shown in Figure 3.

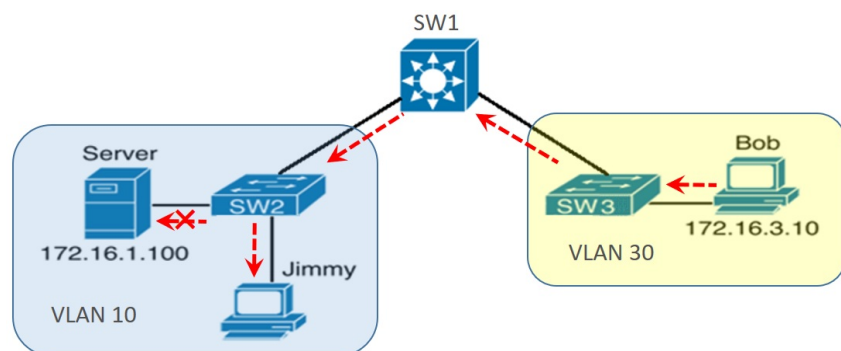


Figure 3

However, this attack is only one-way communication which may alert the victim. To be more damaging, Jimmy may spoof his IP address as well to reply back. In addition, Jimmy may clone the server so that Bob may login and thus be able to steal his credential.

Defending Address Spoofing Attacks

To prevent insider threat actors from conducting IP and MAC address spoofing attacks, **IP source guard** (IPSG) is recommended to be configured at the required interfaces of a switch, which may be enabled to defend against IP address spoofing only, or to defend against both IP and MAC address spoofing.

To enable IPSG against IP address spoofing attack, configure:

```
switch(config)# interface interface-id
switch(config-if)# ip verify source
```

To enable IPSG against both IP and MAC address spoofing attack, configure:

```
switch(config)# interface interface-id
switch(config-if)# ip verify source mac-check
```

Note that **port security** needs to be configured as well for IPSG to be able to defend against MAC address spoofing attack.

To verify whether IPSG is configured correctly, use the show command:

```
switch# show ip verify source
```

Figure 4 shows a sample output when IPSG is enabled against IP address spoofing only.

```
Switch# show ip verify source
Interface  Filter-type  Filter-mode  IP-address  Mac-address  Vlan
-----
Gi0/1     ip           active       40.1.1.24   -            1
```

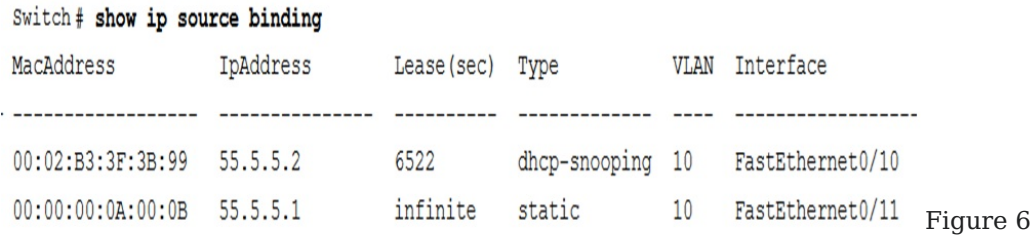
Figure 4

Figure 5 shows a sample output when IPSG is enabled against IP and MAC address spoofing.

```
Switch# show ip verify source
Interface  Filter-type  Filter-mode  IP-address  Mac-address  Vlan
-----
Gi0/1     ip-mac      active       40.1.1.24   00:00:00:00:03:04  1
```

Figure 5

In fact when enabled with IPSG, the switch will maintain a **source binding table** on the IP and MAC addresses permitted on each interface enabled with IPSG. An example is shown in Figure 6.



The source binding table is updated automatically by DHCP snooping binding table if DHCP snooping is enabled, shown as type dhcp-snooping in Figure 6. Otherwise, manual configuration shown as type static in Figure 6 is needed to insert IP and MAC addresses of the required interfaces into the source binding table as follows:

```
switch(config)# ip source binding mac-addr vlan vlan-id ip-addr interface int-id
```

With the availability of source binding table, an interface only permits network traffic with valid source IP and MAC addresses, thus preventing a potential insider threat actor from conducting IP and MAC address spoofing attacks.

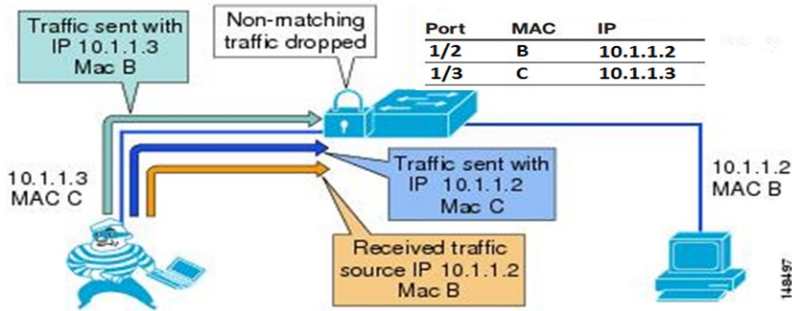


Figure 7

For example in Figure 7, the interface where the threat actor is connected to only permits network traffic with source IP 10.1.1.3 and MAC C. Any spoof address not matching the entries in the source binding table will be dropped.